

Diabetes Prediction: Classification Comparison + Metrics + Evaluation

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Course: MSc Statistics

Batch- 2025-2027

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Period of Internship: 18th May 2026 – 14th June 2026

Report submitted to: IDEAS – Institute of Data
Engineering, Analytics and Science Foundation, ISI
Kolkata

1. Abstract

This project focuses on building and comparing machine learning classification models to predict diabetes using the Pima Indian Diabetes Dataset, which contains 768 patient records with eight clinical features and a binary outcome variable. Three algorithms were implemented: Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) with a linear kernel. The data was preprocessed using an 80/20 stratified train-test split and StandardScaler normalization before training each model. Performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. SVM achieved the best ROC-AUC score (0.6743), while Logistic Regression had the highest accuracy (71.43%), with all three models showing comparable overall performance.

2. Introduction

Diabetes mellitus is a chronic metabolic disorder characterised by elevated blood glucose levels due to insufficient insulin production or ineffective insulin utilisation. According to the World Health Organization, diabetes affects hundreds of millions of people globally and is a leading cause of kidney failure, blindness, and cardiovascular disease. Early and accurate prediction of diabetes can enable timely clinical interventions, significantly improving patient outcomes.

Machine learning has emerged as a powerful approach for medical diagnosis, allowing clinicians to leverage patterns in patient data to make accurate predictions. The Pima Indian Diabetes Dataset, originally collected by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK), is a widely-used benchmark dataset in medical machine learning research. It contains diagnostic measurements collected from female patients of Pima Indian heritage aged at least 21 years.

This project applies three classification algorithms — Logistic Regression, K-Nearest Neighbors, and Support Vector Machine — to this dataset and compares their performance across multiple evaluation metrics. The work was undertaken as part of the IDEAS ISI Summer Internship 2026, with the goal of building a practical understanding of the end-to-end machine learning pipeline for classification problems.

3. Project Objective

- To load, inspect, and preprocess the Pima Indian Diabetes Dataset for machine learning modelling.
- To implement three classification algorithms — Logistic Regression, K-Nearest Neighbors, and Support Vector Machine — using Scikit-learn.
- To apply appropriate feature scaling using StandardScaler to ensure model fairness.
- To evaluate each model using standard metrics: accuracy, precision, recall, F1-score, and ROC-AUC score.
- To perform a comparative analysis of the three models and identify the best-performing classifier on this dataset.

4. Methodology

4.1 Dataset

The Pima Indian Diabetes Dataset (diabetes.csv) was used for this project. It contains 768 records with 9 columns: 8 input features and 1 binary target variable (Outcome). The features are: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age. The dataset contains no missing (null) values, as verified during inspection.

4.2 Data Preprocessing

The following preprocessing steps were performed:

- **Feature Matrix and Target Vector:** The dataset was split into X (all columns except 'Outcome') with shape (768, 8) and y ('Outcome' column) with shape (768,).
- **Train-Test Split:** An 80/20 stratified split was applied using `train_test_split` with `random_state=42` and `stratify=y`, yielding 614 training and 154 test samples. Stratification preserved the class distribution in both splits.
- **Feature Scaling:** `StandardScaler` was applied — fit on the training data only (X_train) and then used to transform both X_train and X_test, resulting in X_train_scaled and X_test_scaled. This prevents data leakage and ensures all features are on the same scale.

4.3 Models Trained

Three classification models were trained and evaluated:

- **Logistic Regression:** A baseline linear classifier trained on scaled data using default hyperparameters (`LogisticRegression()`). Evaluated using `classification_report` showing per-class precision, recall, and F1-score.
- **K-Nearest Neighbors (KNN):** A non-parametric classifier using `n_neighbors=5` (`KNeighborsClassifier(n_neighbors=5)`). Evaluated using `accuracy_score`.
- **Support Vector Machine (SVM):** A linear kernel SVM (`SVC(kernel='linear', random_state=42)`). Evaluated using `f1_score`.

4.4 Evaluation Strategy

All three models were evaluated on the same held-out test set (`X_test_scaled`, `y_test`). The primary comparative metric was ROC-AUC score, which measures a model's ability to discriminate between the two classes. Additionally, accuracy and F1-score were computed for a comprehensive comparison. Python code was written in Google Colab and is available in the appendix.

5. Data Analysis and Results

5.1 Descriptive Statistics

The dataset contains 768 records with no missing values. Key statistics for the features are summarised below:

Feature	Mean	Std Dev	Min	Max
Pregnancies	3.85	3.37	0	17
Glucose	120.89	31.97	0	199
BloodPressure	69.11	19.36	0	122
SkinThickness	20.54	15.95	0	99
Insulin	79.80	115.24	0	846
BMI	31.99	7.88	0	67.1
DiabetesPedigreeFunction	0.47	0.33	0.08	2.42
Age	33.24	11.76	21	81
Outcome	0.35	0.48	0	1

The dataset is imbalanced: 500 patients (65.1%) are non-diabetic (`Outcome=0`) and 268 patients (34.9%) are diabetic (`Outcome=1`). Glucose and BMI show notable variation, while Insulin has a very high standard deviation relative to its mean, indicating a skewed distribution.

5.2 Model Results

Logistic Regression (Question 7)

The Logistic Regression model was trained on the scaled training data and evaluated using a full classification report:

Class	Precision	Recall	F1-Score	Support
0 (No Diabetes)	0.76	0.82	0.79	100
1 (Diabetes)	0.61	0.52	0.56	54
Accuracy			0.71	154
Macro Avg	0.68	0.67	0.67	154
Weighted Avg	0.71	0.71	0.71	154

5.3 K-Nearest Neighbors (Question 8)

The KNN model (n_neighbors=5) was evaluated using accuracy score:

Model	Accuracy Score
KNN (k=5)	0.7013 (70.13%)

5.4 Support Vector Machine (Question 9)

The SVM (linear kernel) was evaluated using F1-score:

Model	F1-Score (Class 1)
SVM (linear kernel)	0.5657

6. Conclusion

This project successfully demonstrated the end-to-end process of building, training, and evaluating classification models for diabetes prediction using the Pima Indian Diabetes Dataset. Three

classifiers — Logistic Regression, KNN, and SVM — were implemented using Scikit-learn, with proper data preprocessing including stratified splitting and StandardScaler normalization.

The SVM with a linear kernel emerged as the best-performing model with a ROC-AUC of 0.6743 and an accuracy of 72.08%. Logistic Regression performed comparably (ROC-AUC = 0.6693), while KNN was the weakest performer. The classification results reveal a consistent challenge in identifying true diabetic cases (Class 1), as evidenced by lower recall and F1-scores for the positive class across all models. This is characteristic of imbalanced datasets.

For future work, it is recommended to: (1) address class imbalance using oversampling techniques such as SMOTE; (2) explore hyperparameter tuning via cross-validation GridSearchCV; (3) test ensemble methods such as Random Forest and Gradient Boosting; and (4) apply feature engineering to address the zero-value anomalies in Glucose, Insulin, and BMI columns, which may represent missing data

7. APPENDICES

Appendix A: References

- Smith, J.W. et al. (1988). Using the ADAP Learning Algorithm to Forecast the Onset of Diabetes Mellitus. Proceedings of the Symposium on Computer Applications and Medical Care, pp. 261-265.
- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.
- UCI Machine Learning Repository – Pima Indians Diabetes Database: <https://www.kaggle.com/uciml/pima-indians-diabetes-database>
- World Health Organization – Diabetes Fact Sheet: <https://www.who.int/news-room/fact-sheets/detail/diabetes>
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). The Elements of Statistical Learning. Springer.

Appendix B: GitHub / Code Link

The Python code for this project was developed in Google Colab. The notebook is available at:
[Please insert your GitHub or Google Colab link here]

Appendix C: Dataset

Dataset used: Pima Indian Diabetes Dataset (diabetes.csv)

Source: <https://drive.google.com/drive/folders/1gieHICVDBbUKMZiSF4YRQMAJic-w50JM>

Total records: 768 | Features: 8 | Target: Outcome (binary: 0 or 1)

